

## Ethernet Standard And Specification Pending

1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

ANS :

| Pin | T568A Color  | T568B Color  |
|-----|--------------|--------------|
| 1   | White/Green  | White/Orange |
| 2   | Green        | Orange       |
| 3   | White/Orange | White/Green  |
| 4   | Blue         | Blue         |
| 5   | White/Blue   | White/Blue   |
| 6   | Orange       | Green        |
| 7   | White/Brown  | White/Brown  |
| 8   | Brown        | Brown        |

RJ-45 Connector (Front View)

-----  
 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |  
 -----

2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

ANS :

| Standard   | Speed              | Cable Type | Maximum Length | Connector |
|------------|--------------------|------------|----------------|-----------|
| 10BASE-T   | 10 Mbps            | Cat3/Cat5  | 100 m          | RJ-45     |
| 100BASE-TX | 100 Mbps           | Cat5       | 100 m          | RJ-45     |
| 1000BASE-T | 1000 Mbps (1 Gbps) | Cat5e/Cat6 | 100 m          | RJ-45     |

3. **Given a scenario** where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why. Think about how transmit and receive pairs are wired in each cable type.

**ANS :**

Cable Used: Crossover Cable

Reason:

A crossover cable swaps the transmit (TX) and receive (RX) wire pairs, allowing two similar devices (laptop-to-laptop) to communicate directly without a switch or hub.

4. **List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.**

**ANS :**

| Device                          | Likely Ethernet Standard |
|---------------------------------|--------------------------|
| WiFi Router                     | 1000BASE-T               |
| sDesktop Computer (College Lab) | 1000BASE-T               |
| Network Printer                 | 100BASE-TX               |

Explanation:

- WiFi Router: Uses Gigabit Ethernet for high-speed internet.
- Desktop Computer: Usually supports Gigabit Ethernet for fast network access.
- Network Printer: Often uses 100 Mbps Ethernet, which is sufficient for printing tasks.